Microsoft System Design Interview Guide

This folder contains **52 system design questions** commonly asked at Microsoft interviews, each in its own dedicated guide.

📚 Questions by Category

Microsoft Products & Azure Services (1-10, 31-36, 48-51)

#	Topic	File
1	Microsoft Teams (Chat, Video, Presence)	01-microsoft-teams.md
2	Scalable Azure Service	02-scalable-azure-service.md
3	Multi-Tenant Cloud Solution	03-architect-a-multi-tenant-cloud-solution.md
10	Word Online (Collaborative Editing)	10-design-collaborative-document-editing.md
12	OneDrive (File Storage)	12-design-distributed-file-storage.md
31	Teams Video Conferencing	31-design-teams-video-conferencing.md
32	Xbox Live Leaderboard	32-design-xbox-live-leaderboard.md
33	Xbox Matchmaking	33-design-xbox-matchmaking.md
34	LinkedIn News Feed	34-design-linkedin-news-feed.md
35	LinkedIn People You May Know	35-design-linkedin-people-you-may-know.md
36	Azure Key Vault	36-design-azure-key-vault.md
48	**Azure Entra Identity Platform**	48-design-azure-entra-identity-platform.md
49	**GitHub Copilot**	49-design-github-copilot.md
50	**Azure AI Foundry**	50-design-azure-ai-foundry.md
51	**AI-Powered SOC**	51-design-ai-soc-platform.md

Infrastructure & Scalability (11-30, 37-41)

#	Topic	File
7	Content Delivery Network (CDN)	07-design-a-content-delivery-network.md
11	Notification Service	11-design-a-notification-service.md
13	Cost Optimization	13-optimize-cost-while-maintaining-performance.md
14	High Availability (Multi-Region)	14-design-high-availability-across-regions.md
16	Rate Limiter	16-design-a-rate-limiter.md
17	Sharding & Partitioning	17-implement-sharding-and-partitioning.md
22	Logging & Monitoring	22-design-logging-monitoring.md
23	Load Balancer	23-design-a-load-balancer.md
24	Eventual Consistency	24-handle-eventual-consistency.md
25	Real-Time Log Processing	25-design-real-time-log-processing.md
26	CAP Theorem Trade-offs	26-handle-cap-theorem-trade-offs.md
27	Microservices vs Monolith	27-microservices-vs-monolith.md
28	Database Schema Evolution	28-database-schema-evolution.md
37	Distributed ID Generator	37-design-distributed-id-generator.md
38	Distributed Cache	38-design-distributed-cache.md
39	Distributed Job Scheduler	39-design-distributed-job-scheduler.md
40	Key-Value Store	40-design-key-value-store.md
41	Distributed Locking Service	41-design-distributed-locking-service.md

Web Services & Applications (4-9, 15, 18-21, 29-30, 42-47, 52)

#	Topic	File
4	URL Shortener	04-design-a-url-shortening-service.md
5	Search Autocomplete/Typeahead	05-design-search-autocomplete-typeahead.md

6	Uber/Lyft Ride-Sharing	06-design-uber-lyft-ride-sharing-platform.md
8	WhatsApp/Slack Chat	08-design-whatsapp-slack-chat-messaging.md
9	Web Crawler	09-design-a-scalable-web-crawler.md
15	Meeting Scheduling System	15-design-meeting-scheduling-system.md
18	Video Streaming (YouTube/Netflix)	18-design-video-streaming-platform.md
19	Message Ordering in Chat	19-ensure-message-ordering-in-chat.md
20	Data Security (GDPR/HIPAA)	20-secure-sensitive-data-gdpr-hipaa.md
21	Real-Time Location Tracking	21-design-real-time-location-tracking.md
29	Recommendation Engine	29-design-a-recommendation-engine.md
30	Trade-off Discussions	30-trade-off-discussions.md
42	Ticket Booking (Ticketmaster)	42-design-ticket-booking-system.md
43	Proximity Service (Yelp/Maps)	43-design-proximity-service.md
44	Web Analytics	44-design-web-analytics-system.md
45	Ad Bidding System	45-design-ad-bidding-system.md
46	E-Commerce Platform	46-design-e-commerce-platform.md
47	Payment Gateway	47-design-payment-gateway.md
52	Twitter/X Social Platform	52-design-twitter.md

📖 Guide Structure

Each guide follows a consistent format:

1. **Problem Statement** - What you're designing
2. **Clarifying Questions** - Questions to ask the interviewer
3. **High-Level Design** - ASCII architecture diagram
4. **Key Components** - Table of components and technologies
5. **Scalability Considerations** - How to scale
6. **Trade-offs** - Design decisions with pros/cons
7. **How to Explain It Well** - Interview talking points

🎯 Key Patterns to Know

Pattern	Used In
CAP Theorem	#26, #24, #14
Consistent Hashing	#17, #38, #40
Saga Pattern	#47, #42
CQRS	#34, #29
Event Sourcing	#19, #25
Pub/Sub	#1, #8, #11
SFU (Video)	#1, #31
Geohashing	#6, #21, #43
Trie	#5
LSM Tree	#40
OAuth/OIDC	#48
Zero Trust	#48
Fill-in-the-Middle (FIM)	#49
Speculative Decoding	#49
GPU Inference Optimization	#49
Model Router	#50
Hub/Project Isolation	#50
LoRA Fine-Tuning	#50
ReAct Agent Loop	#51
UEBA / Entity Risk Scoring	#51
KQL / Stream Processing	#51
Fan-out on Write	#52
Celebrity/Hybrid Fan-out	#52

🚀 Quick Start

1. **Start with Microsoft Products** (#1, #10, #31-36, #48-51) - Most likely to be asked
2. **Master Infrastructure** (#16, #17, #23, #37-41) - Foundations for all designs
3. **Practice Trade-offs** (#26, #27, #30) - Critical for L5+ interviews
4. **Know Azure Identity** (#48) - Essential for any Azure-based design

Good luck with your interviews! 🎉